UCL DATA ENGINEERING GROUP ASSIGNMENT

Cyber Risk Dashboard

A static GitHub Pages dashboard built on transformed vulnerability datasets from NVD, CISA KEV, EPSS, and cvelistV5.

CVE Rows	KEV Rate	High EPSS Groups
5,133	0.4%	2

Overview

Key summary metrics generated from the current transformed datasets.

TOTAL CVES
5,133
Rows in the current vulnerability mart

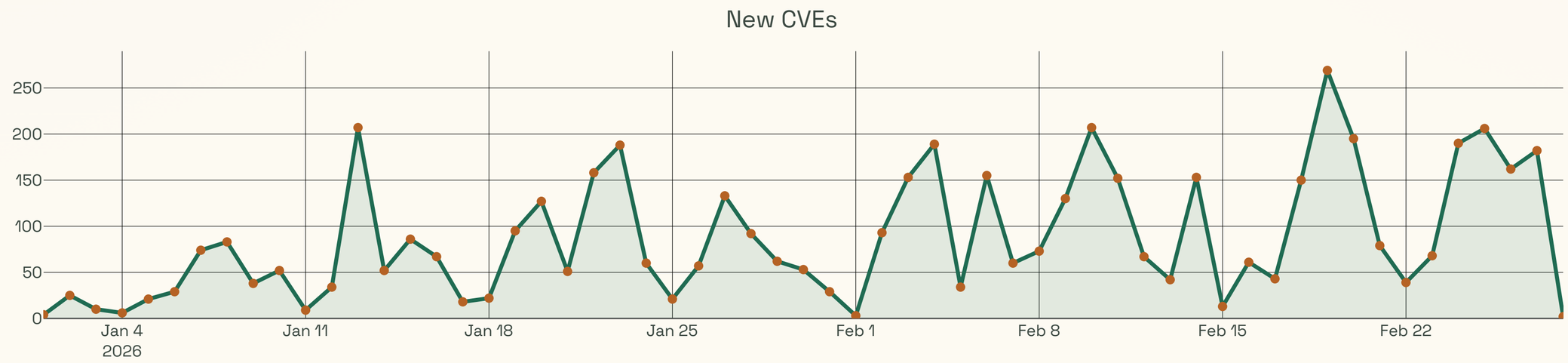
HIGHEST-VOLUME PRODUCT
microsoft / windows_10_1809
171 CVEs in this group

TOP AVERAGE EPSS
gnu / inetutils
Average EPSS 0.7792

MOST FREQUENT CWE GROUP
No CWE | samsung / android
795 CVEs in this grouped pattern

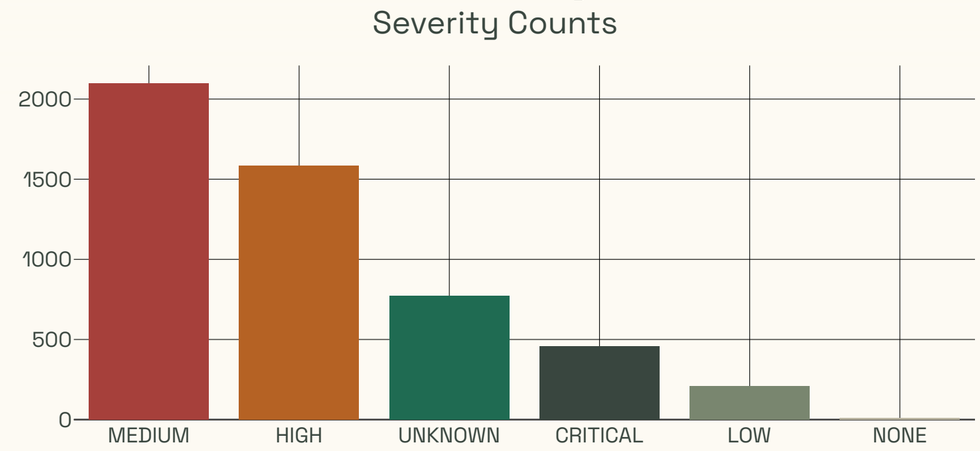
Daily New CVEs

Published CVE counts over time.



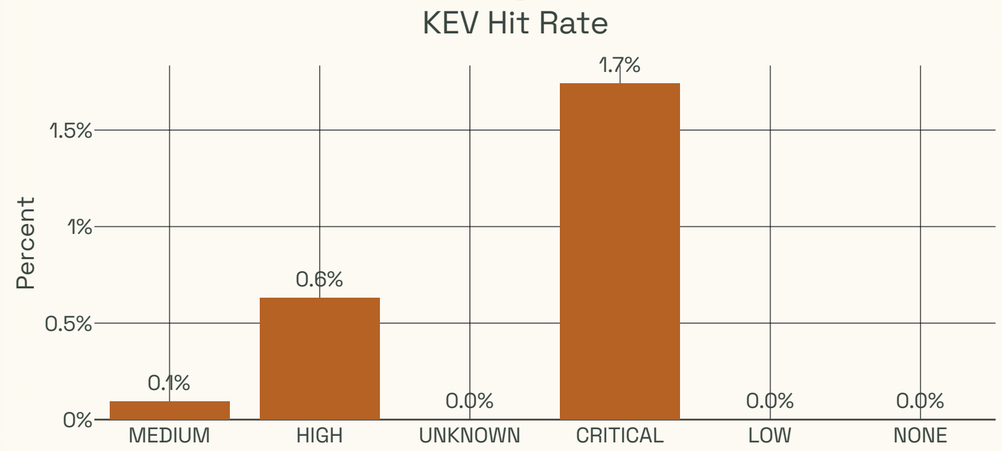
CVSS Severity Distribution

How vulnerabilities are distributed across severity bands.



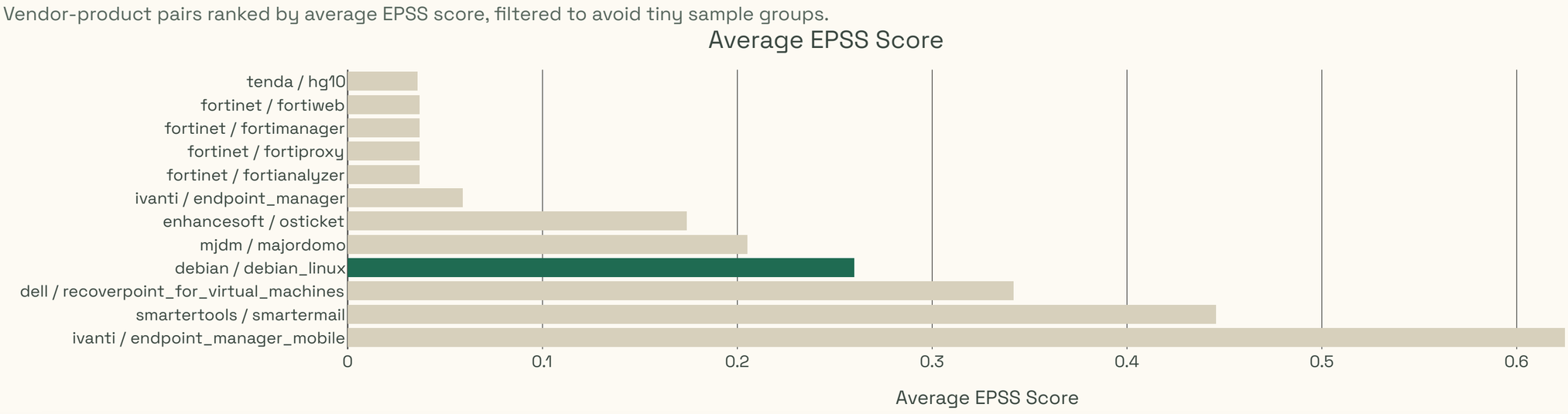
KEV Hit Rate by Severity

Share of CVEs in KEV for each CVSS severity level.



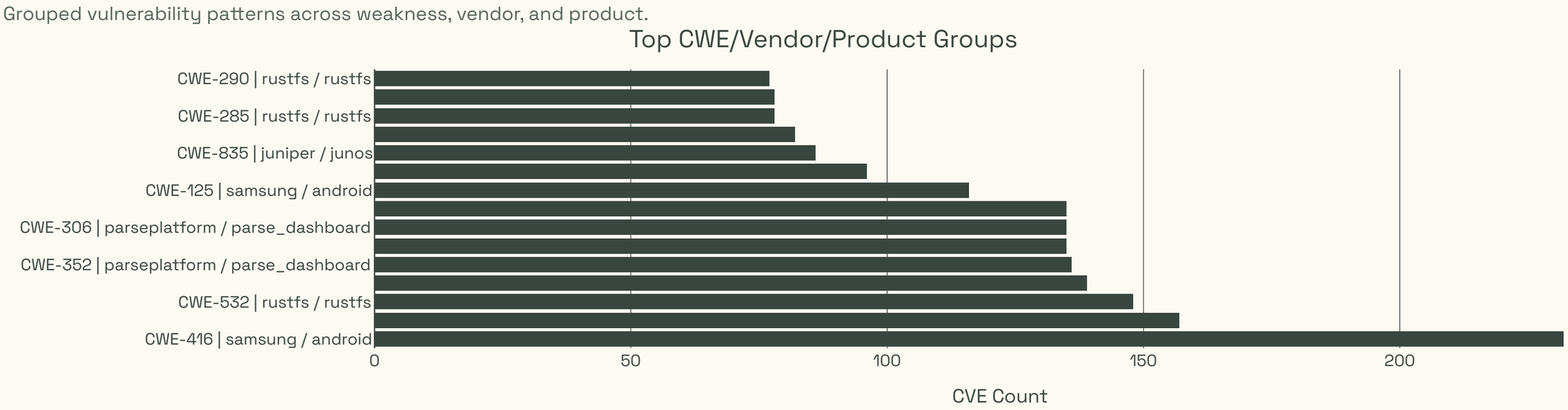
Top Vendor/Product Average EPSS Groups

Vendor-product pairs ranked by average EPSS score, filtered to avoid tiny sample groups.



Top CWE, Vendor, Product Aggregations

Grouped vulnerability patterns across weakness, vendor, and product.



Methodology

How the dashboard is produced from the project pipeline.

This dashboard is built on top of transformed summary datasets generated from four public sources: NVD, CISA KEV, EPSS, and cvelistV5.

The pipeline first lands raw source files, then creates a unified vulnerability mart, product bridge tables, official CVE enrichment tables, and reporting summaries.

Product-level charts use the CVE-product bridge layer rather than only the flat master table, so the visualisations better reflect many-to-many relationships between CVEs and affected products.

Data can be loaded either from static CSV files under `data/curated/transformation_summaries` or from the FastAPI service using `?api_base=http://localhost:8001`.